Forecasting On Air Passenger data

Project Pipeline

Load Data ↓ Clean & Handle Missing Values ↓ Convert Date Index ↓ Visualization ↓ Stationarity Check ↓ Differencing ↓ Train/Test Split ↓ Model Training (ARIMA) ↓ Forecast ↓ Evaluation ↓ Future Prediction

In [55]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

Load AirPassengers Dataset

In [56]:
```
df=pd.read_csv(r'C:\Users\anupa\Desktop\Python Learning\Forecasting\AirPassenger
```

In [57]:
```
df.head()
```

Out[57]:

	Month	Passengers
0	1949-01	112
1	1949-02	118
2	1949-03	132
3	1949-04	129
4	1949-05	121

In [58]:
```
df.tail()
```

Out[58]:

	Month	Passengers
139	1960-08	606
140	1960-09	508
141	1960-10	461
142	1960-11	390
143	1960-12	432

In [59]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Month       144 non-null    object
 1   Passengers  144 non-null    int64
dtypes: int64(1), object(1)
memory usage: 2.4+ KB
```

Convert Month Column

In [60]:
```
df['Month'] = pd.to_datetime(df['Month'])
```

In [61]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Month       144 non-null    datetime64[ns]
 1   Passengers  144 non-null    int64
dtypes: datetime64[ns](1), int64(1)
memory usage: 2.4 KB
```

In [62]:
```
df.set_index('Month', inplace=True)
```

Check Missing Values

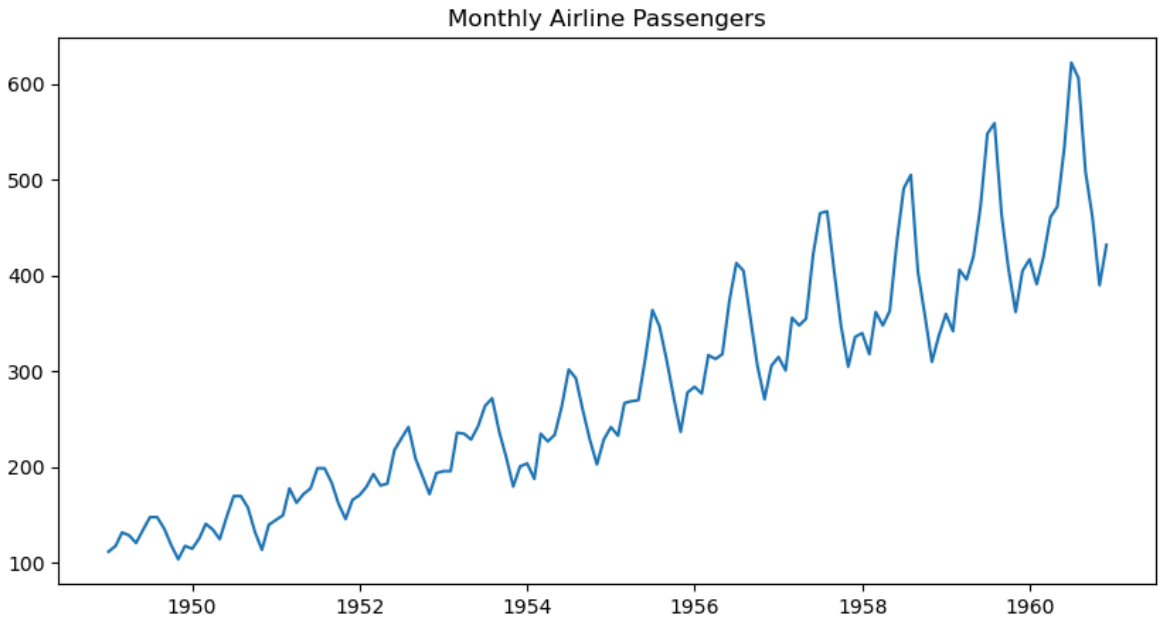
In [63]:
```
df.isnull().sum()
```

Out[63]:
```
Passengers    0
dtype: int64
```

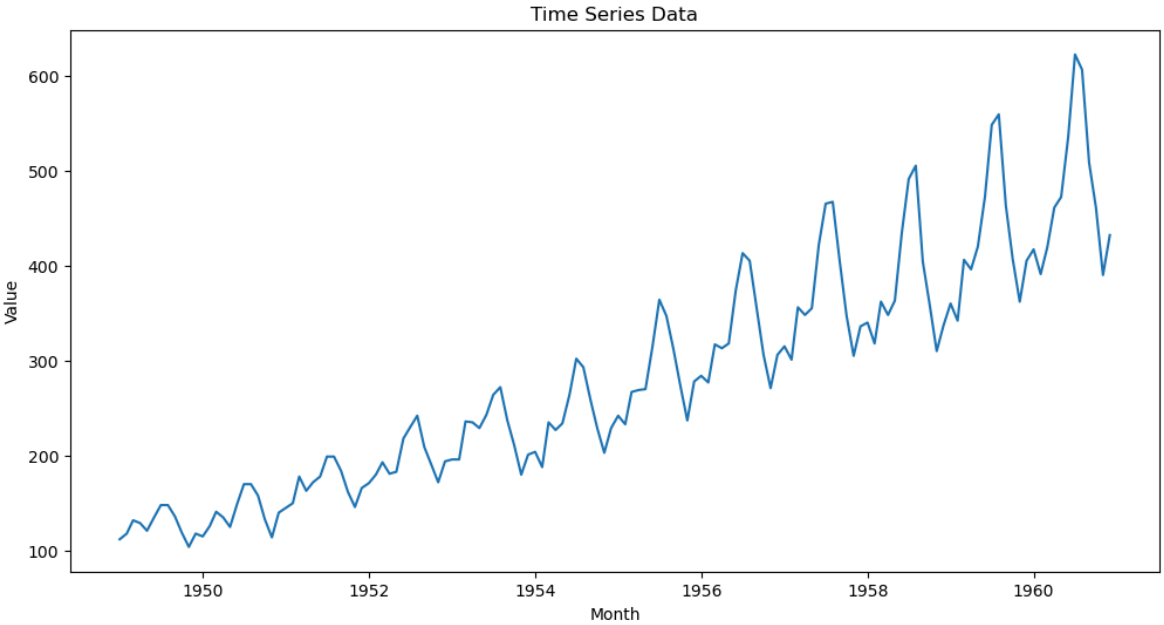
In [64]:
```
df = df.dropna()
```

Visualize Time Series

In [65]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))
plt.plot(df)
plt.title("Monthly Airline Passengers")
plt.show()
```



In [66]:
```
plt.figure(figsize=(12,6))
plt.plot(df)
plt.title("Time Series Data")
plt.xlabel("Month")
plt.ylabel("Value")
plt.show()
```



Check Stationarity

In [67]:
```
result = adfuller(df['Passengers'])

print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: 0.8153688792060511
p-value: 0.991880243437641
```

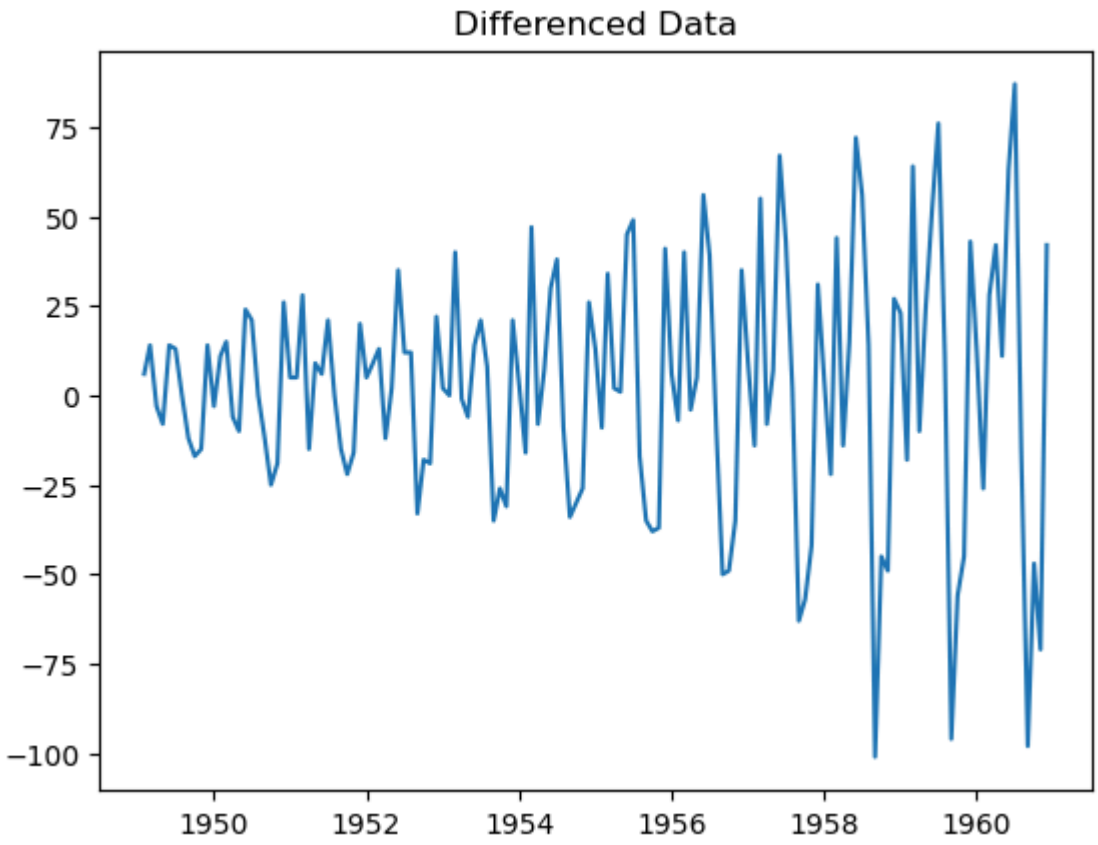
Interpretation:

p < 0.05 → Stationary

p > 0.05 → Need differencing

Difference the Series

In [68]:
```
df_diff = df.diff().dropna()

plt.plot(df_diff)
plt.title("Differenced Data")
plt.show()
```



Second Stationarity Checking

In [69]:
```
result = adfuller(df_diff['Passengers'])

print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

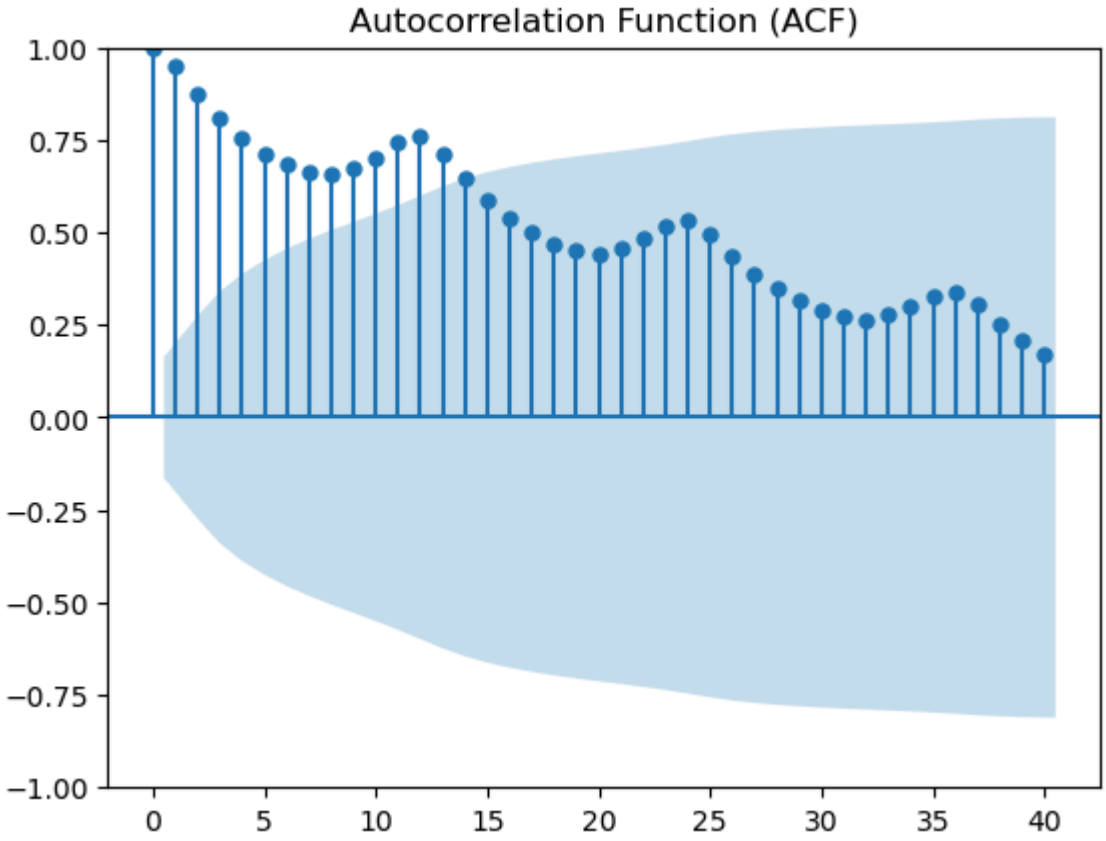
```
ADF Statistic: -2.8292668241699874
p-value: 0.05421329028382711
```

In [70]:
```
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
```

ACF PLot

In [71]:
```
plt.figure(figsize=(10,5))

plot_acf(df['Passengers'], lags=40)

plt.title("Autocorrelation Function (ACF)")
plt.show()
```

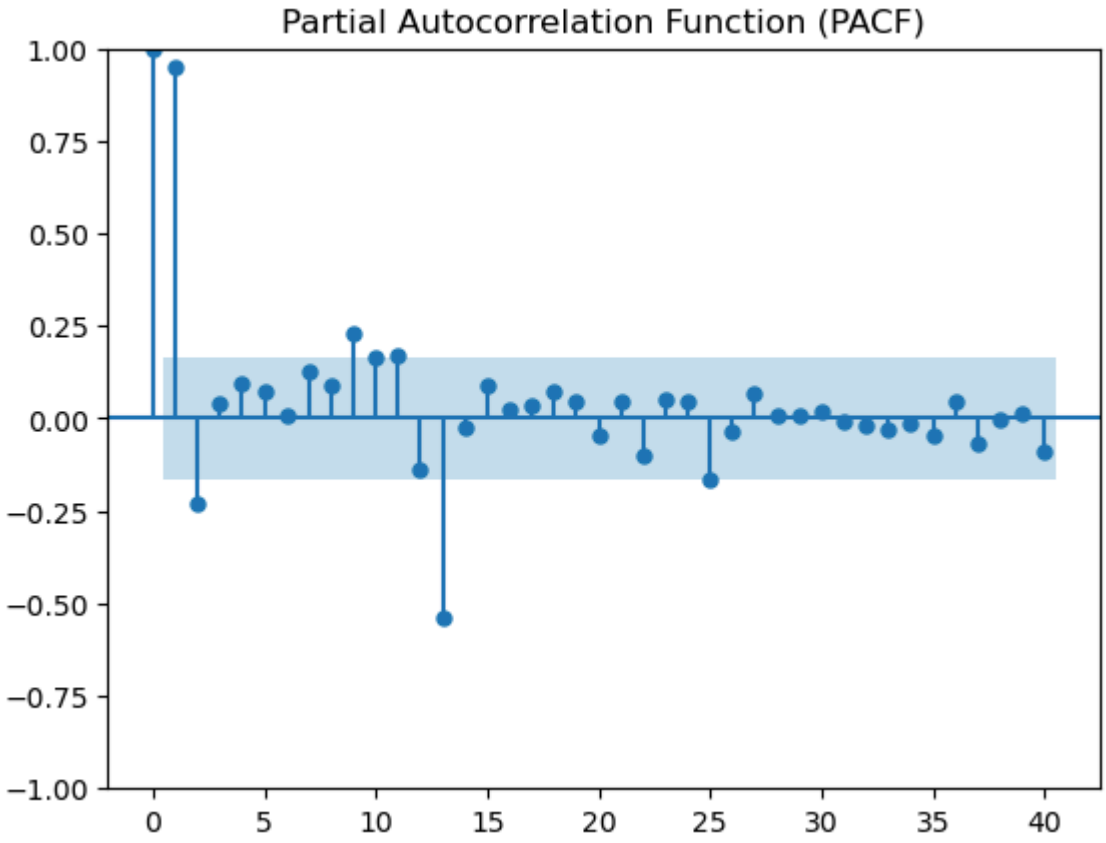
```
<Figure size 1000x500 with 0 Axes>
```



Pacf Plat

In [72]:
```
plt.figure(figsize=(10,5))

plot_pacf(df['Passengers'], lags=40)

plt.title("Partial Autocorrelation Function (PACF)")
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```



Train Test Split

In [73]:
```
train_size = int(len(df) * 0.8)

train = df[:train_size]
test = df[train_size:]
```

Build ARIMA Model

In [74]:
```
model = ARIMA(train['Passengers'], order=(4,1,12))
model_fit = model.fit()
print(model_fit.summary())
```

```
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency MS wil
l be used.
  self._init_dates(dates, freq)
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency MS wil
l be used.
  self._init_dates(dates, freq)
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency MS wil
l be used.
  self._init_dates(dates, freq)
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\statespace\sarimax.py:
978: UserWarning: Non-invertible starting MA parameters found. Using zeros as sta
rting parameters.
  warn('Non-invertible starting MA parameters found.'
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:             Passengers   No. Observations:                  115
Model:                ARIMA(4, 1, 12)   Log Likelihood                -476.574
Date:                Thu, 12 Mar 2026   AIC                            987.148
Time:                        09:59:38   BIC                           1033.664
Sample:                    01-01-1949   HQIC                          1006.026
                         - 07-01-1958
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          1.6633      0.070     23.864      0.000       1.527       1.800
ar.L2         -1.8088      0.127    -14.231      0.000      -2.058      -1.560
ar.L3          1.5601      0.127     12.307      0.000       1.312       1.809
ar.L4         -0.9266      0.059    -15.829      0.000      -1.041      -0.812
ma.L1         -1.7062      0.607     -2.813      0.005      -2.895      -0.517
ma.L2          1.5794      1.788      0.883      0.377      -1.925       5.084
ma.L3         -1.4709      4.625     -0.318      0.750     -10.535       7.594
ma.L4          0.8814      3.186      0.277      0.782      -5.362       7.125
ma.L5          0.1661      0.647      0.257      0.797      -1.102       1.434
ma.L6         -0.0494      1.169     -0.042      0.966      -2.340       2.242
ma.L7          0.1753      0.729      0.240      0.810      -1.254       1.605
ma.L8         -1.0306      0.996     -1.035      0.301      -2.982       0.921
ma.L9          1.3238      0.971      1.363      0.173      -0.580       3.228
ma.L10        -1.4271      2.684     -0.532      0.595      -6.687       3.833
ma.L11         1.5528      4.238      0.366      0.714      -6.754       9.860
ma.L12        -0.6597      1.700     -0.388      0.698      -3.991       2.672
sigma2       192.6943    489.420      0.394      0.694    -766.551    1151.939
===================================================================================
==
Ljung-Box (L1) (Q):                   2.95   Jarque-Bera (JB):                 1.
90
Prob(Q):                              0.09   Prob(JB):                         0.
39
Heteroskedasticity (H):               4.44   Skew:                             0.
31
Prob(H) (two-sided):                  0.00   Kurtosis:                         2.
92
===================================================================================
==

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-st
ep).
```

```
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\base\model.py:607: Converg
enceWarning: Maximum Likelihood optimization failed to converge. Check mle_retval
s
  warnings.warn("Maximum Likelihood optimization failed to "
```

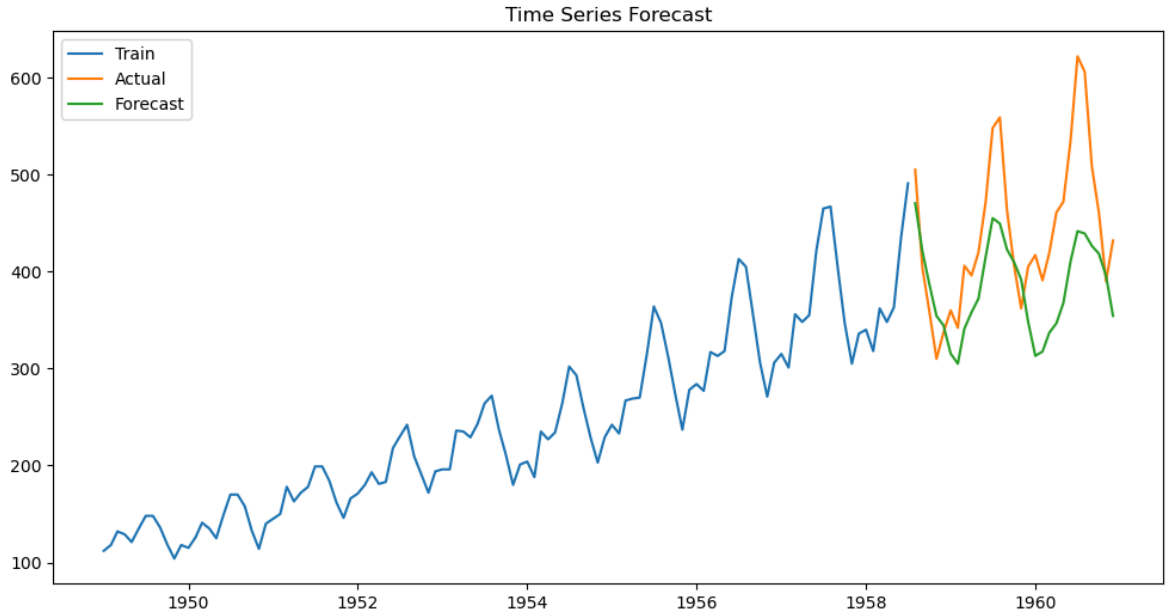
Forecast

In [75]:
```
forecast = model_fit.forecast(steps=len(test))
forecast = pd.Series(forecast, index=test.index)
forecast
```

Out[75]:
```
Month
1958-08-01    470.543158
1958-09-01    421.028322
1958-10-01    387.025526
1958-11-01    354.167550
1958-12-01    344.483601
1959-01-01    315.376145
1959-02-01    304.999112
1959-03-01    341.036868
1959-04-01    357.994043
1959-05-01    372.194188
1959-06-01    415.348540
1959-07-01    455.073694
1959-08-01    449.531771
1959-09-01    422.625184
1959-10-01    409.883699
1959-11-01    391.905139
1959-12-01    348.207297
1960-01-01    313.098103
1960-02-01    317.500037
1960-03-01    336.814279
1960-04-01    346.694255
1960-05-01    367.590790
1960-06-01    410.529709
1960-07-01    441.669344
1960-08-01    439.241252
1960-09-01    426.502512
1960-10-01    418.499676
1960-11-01    395.589064
1960-12-01    354.334062
Name: predicted_mean, dtype: float64
```

Visualization of Forecast

In [76]:
```
plt.figure(figsize=(12,6))

plt.plot(train, label="Train")
plt.plot(test, label="Actual")
plt.plot(forecast, label="Forecast")

plt.legend()
plt.title("Time Series Forecast")
plt.show()
```



Evaluate Model

In [77]:
```
mae = mean_absolute_error(test, forecast)
rmse = np.sqrt(mean_squared_error(test, forecast))
from sklearn.metrics import mean_absolute_percentage_error
forecast = model_fit.predict(start=len(train), end=len(df)-1)
y_true = test['Passengers']
y_pred = forecast
mape = mean_absolute_percentage_error(y_true, y_pred)
print("MAPE:", mape*100)
print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAPE: 14.153692801273564
MAE: 65.85099595444427
RMSE: 79.31109973905562
```

Future Forecast

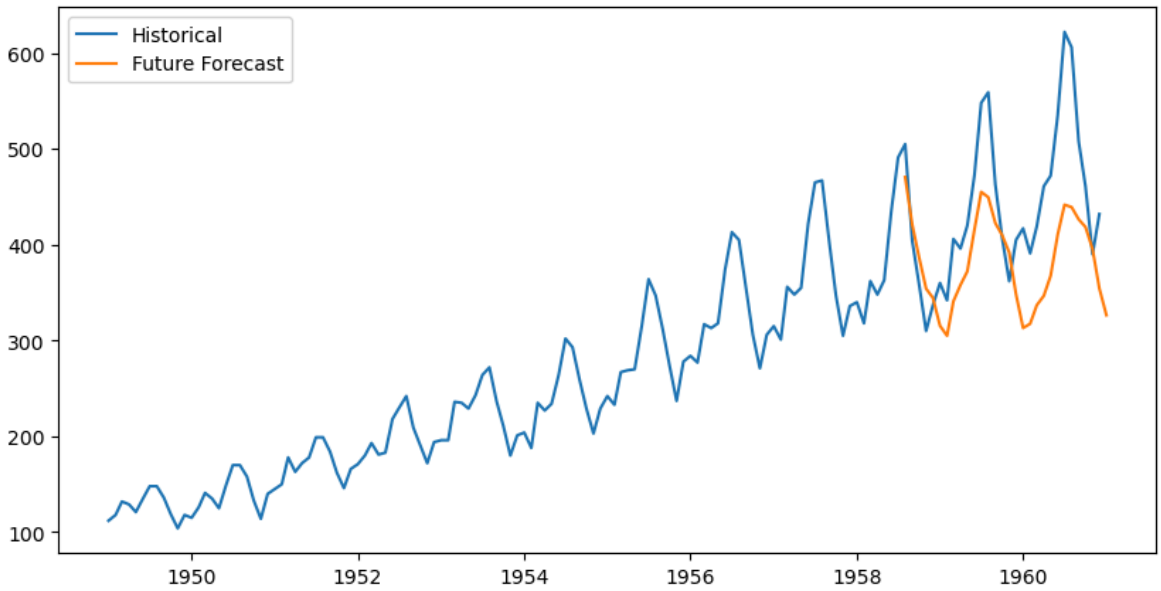
Forecast next 30 days.

In [78]:
```
future = model_fit.forecast(steps=30)

print(future)
```

```
1958-08-01    470.543158
1958-09-01    421.028322
1958-10-01    387.025526
1958-11-01    354.167550
1958-12-01    344.483601
1959-01-01    315.376145
1959-02-01    304.999112
1959-03-01    341.036868
1959-04-01    357.994043
1959-05-01    372.194188
1959-06-01    415.348540
1959-07-01    455.073694
1959-08-01    449.531771
1959-09-01    422.625184
1959-10-01    409.883699
1959-11-01    391.905139
1959-12-01    348.207297
1960-01-01    313.098103
1960-02-01    317.500037
1960-03-01    336.814279
1960-04-01    346.694255
1960-05-01    367.590790
1960-06-01    410.529709
1960-07-01    441.669344
1960-08-01    439.241252
1960-09-01    426.502512
1960-10-01    418.499676
1960-11-01    395.589064
1960-12-01    354.334062
1961-01-01    326.474064
Freq: MS, Name: predicted_mean, dtype: float64
```

Plot future prediction:

In [79]:
```
plt.figure(figsize=(10,5))
plt.plot(df, label="Historical")
plt.plot(future, label="Future Forecast")
plt.legend()
plt.show()
```



Forecasting On WALLMART data

Project Workflow

1 Data Collection (Walmart Dataset) ↓ 2 Data Cleaning & Preprocessing ↓ 3 Exploratory Data Analysis ↓ 4 Feature Aggregation ↓ 5 Train/Test Split ↓ 6 ARIMA Model Training ↓ 7 Forecasting ↓ 8 Model Evaluation ↓ 9 Future Sales Prediction

Load Dataset

In [281…
```
df_1=pd.read_csv(r'C:\Users\anupa\Desktop\Python Learning\Forecasting\train.csv'
```

In [282…
```
df_1.head()
```

Out[282…

	Store	Dept	Date	Weekly_Sales	IsHoliday
0	1	1	2010-02-05	24924.50	False
1	1	1	2010-02-12	46039.49	True
2	1	1	2010-02-19	41595.55	False
3	1	1	2010-02-26	19403.54	False
4	1	1	2010-03-05	21827.90	False

Data Preprocessing

Convert Date column to datetime.

In [283…
```
df_1['Date'] = pd.to_datetime(df_1['Date'])
```

In [284…
```
df_1 =df_1.sort_values('Date')
```

Aggregate Weekly Sales

In [285…
```
sales = df_1.groupby('Date')['Weekly_Sales'].sum().reset_index()
sales.head()
```

Out[285…

	Date	Weekly_Sales
0	2010-02-05	49750740.50
1	2010-02-12	48336677.63
2	2010-02-19	48276993.78
3	2010-02-26	43968571.13
4	2010-03-05	46871470.30

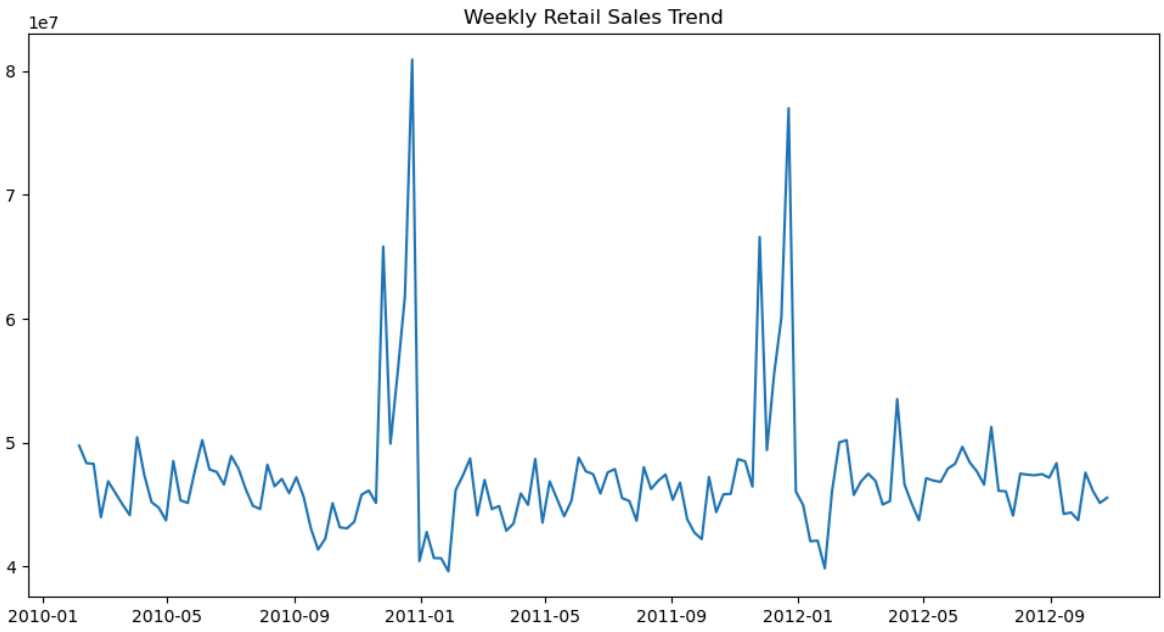
In [286…

```
sales.set_index('Date', inplace=True)
```

Exploratory Data Analysis

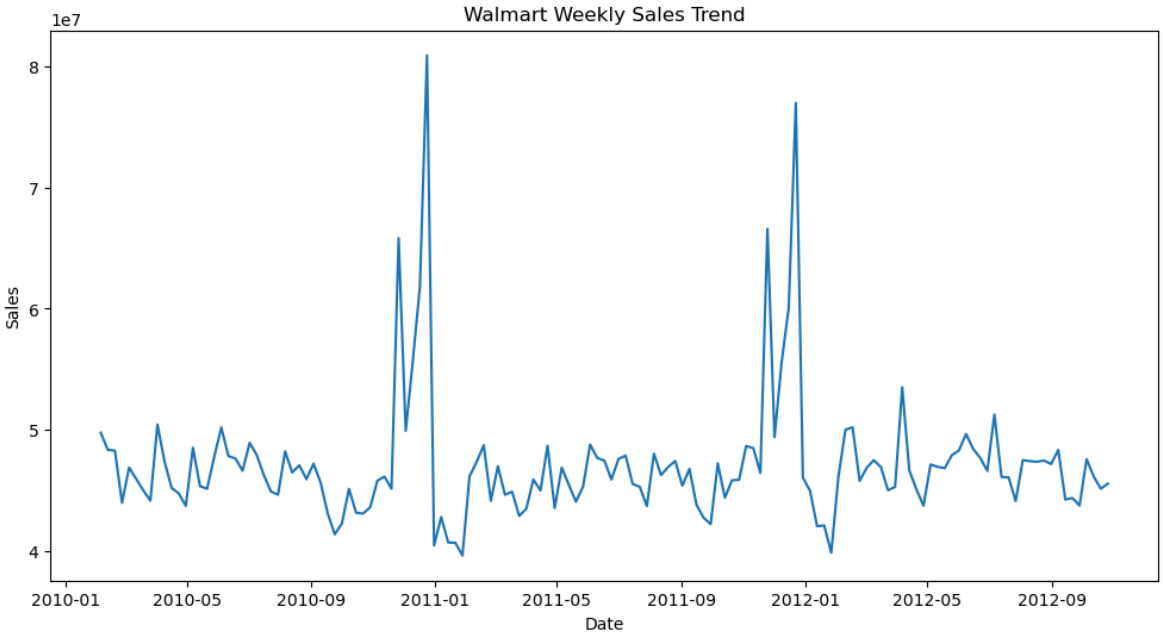
In [287…

```
plt.figure(figsize=(12,6))
plt.plot(sales)
plt.title("Weekly Retail Sales Trend")
plt.show()
```



In [288…

```
plt.figure(figsize=(12,6))
plt.plot(sales)
plt.title("Walmart Weekly Sales Trend")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.show()
```



Train Test Split

In [289…

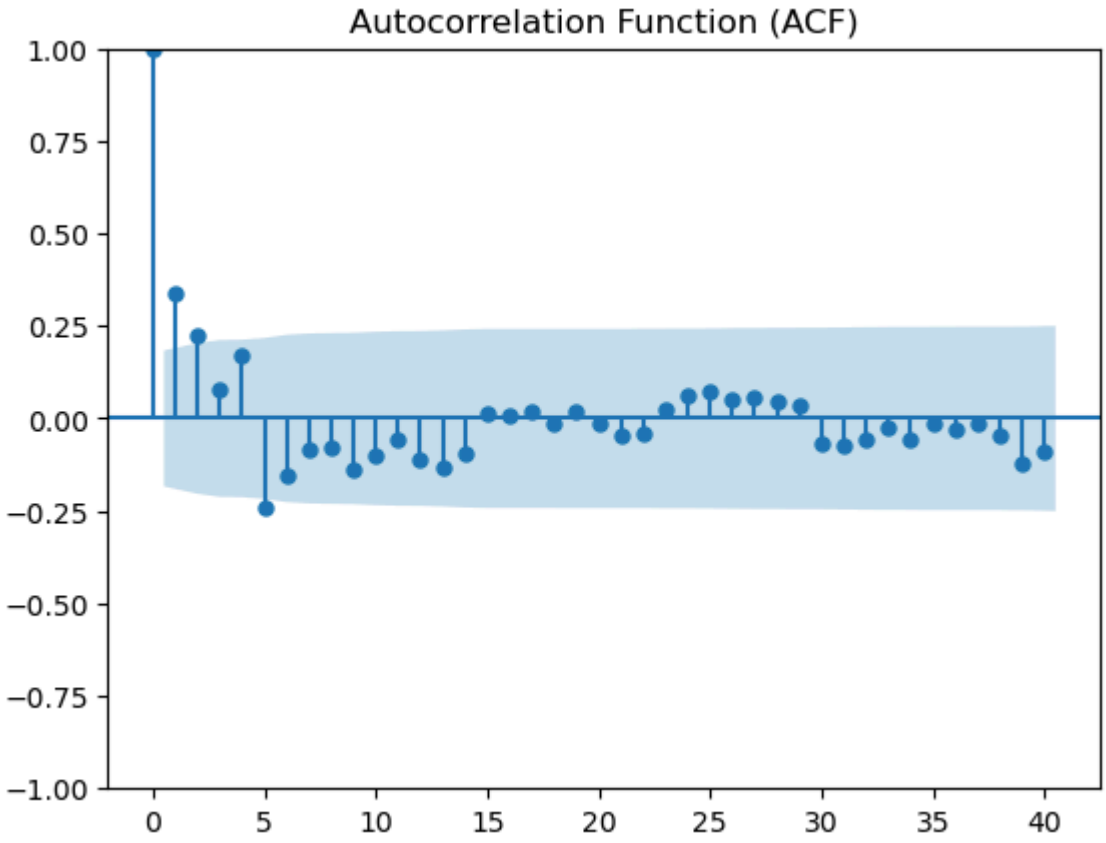
```
train_size = int(len(sales)*0.8)
train = sales[:train_size]
test = sales[train_size:]
```

ACF Plot

In [290…

```
plt.figure(figsize=(10,5))

plot_acf(train['Weekly_Sales'], lags=40)

plt.title("Autocorrelation Function (ACF)")
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```

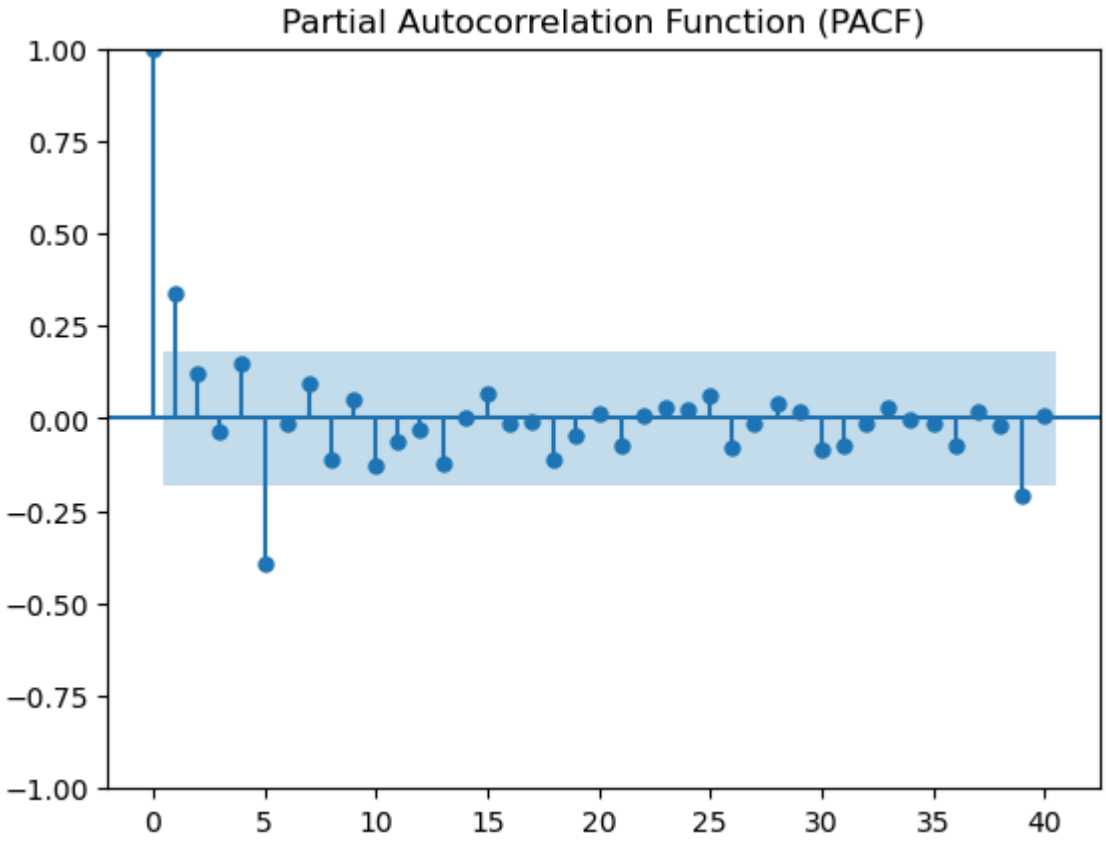


PACF Plot

In [291…

```
plt.figure(figsize=(10,5))

plot_pacf(train['Weekly_Sales'], lags=40)

plt.title("Partial Autocorrelation Function (PACF)")
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```



Build ARIMA Model

In [292…

```
model = ARIMA(train['Weekly_Sales'], order=(5,1,0))
model_fit = model.fit()
print(model_fit.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:           Weekly_Sales   No. Observations:                  114
Model:                 ARIMA(5, 1, 0)   Log Likelihood               -1923.247
Date:                Thu, 12 Mar 2026   AIC                           3858.493
Time:                        10:55:50   BIC                           3874.858
Sample:                    02-05-2010   HQIC                          3865.134
                         - 04-06-2012
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.2956      0.051     -5.829      0.000      -0.395      -0.196
ar.L2         -0.2007      0.059     -3.395      0.001      -0.317      -0.085
ar.L3         -0.1744      0.073     -2.400      0.016      -0.317      -0.032
ar.L4          0.0952      0.072      1.315      0.188      -0.047       0.237
ar.L5         -0.1553      0.070     -2.204      0.028      -0.293      -0.017
sigma2      3.308e+13   3.95e-16   8.37e+28      0.000    3.31e+13    3.31e+13
===================================================================================
==
Ljung-Box (L1) (Q):                   2.59   Jarque-Bera (JB):               194.
61
Prob(Q):                              0.11   Prob(JB):                         0.
00
Heteroskedasticity (H):               3.67   Skew:                            -0.
32
Prob(H) (two-sided):                  0.00   Kurtosis:                         9.
40
===================================================================================
==

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-st
ep).
[2] Covariance matrix is singular or near-singular, with condition number 1.98e+4
4. Standard errors may be unstable.
```

```
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency W-FRI
will be used.
  self._init_dates(dates, freq)
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency W-FRI
will be used.
  self._init_dates(dates, freq)
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency W-FRI
will be used.
  self._init_dates(dates, freq)
```

Forecast Test Data

In [293…

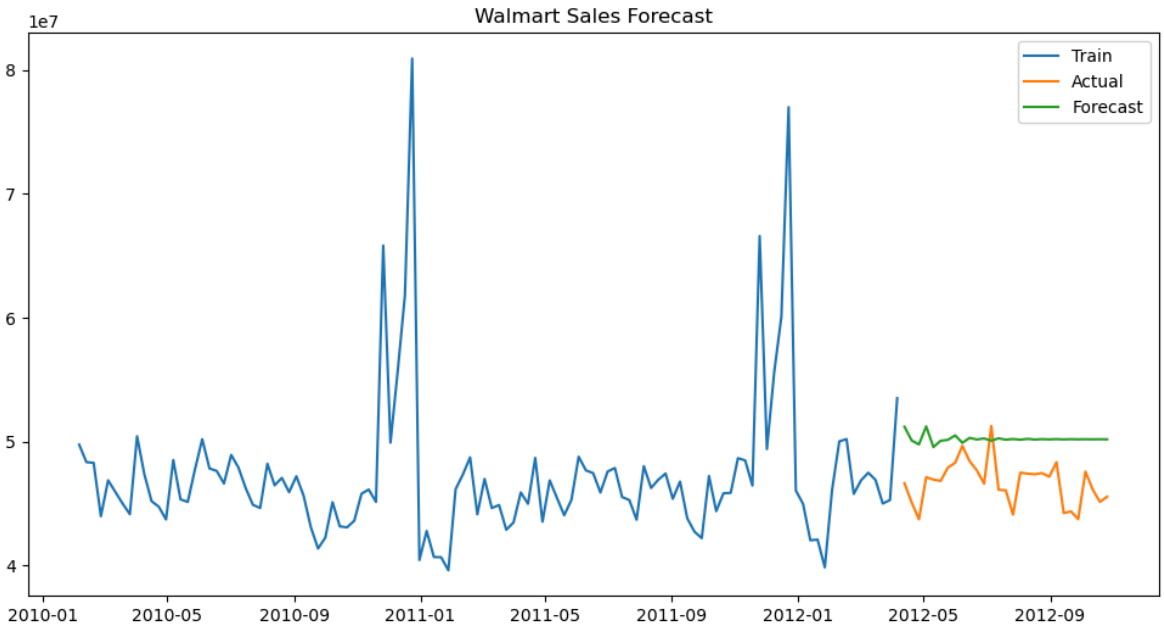
```
forecast = model_fit.forecast(steps=len(test))
forecast = pd.Series(forecast, index=test.index)
print("Test length:", len(test))
print("Forecast length:", len(forecast))
```

```
Test length: 29
Forecast length: 29
```

Visualization

In [294…

```
plt.figure(figsize=(12,6))

plt.plot(train, label="Train")
plt.plot(test, label="Actual")
plt.plot(forecast, label="Forecast")

plt.legend()
plt.title("Walmart Sales Forecast")
plt.show()
```



Model Evaluation

In [295…

```
mae = mean_absolute_error(test, forecast)
rmse = np.sqrt(mean_squared_error(test, forecast))
from sklearn.metrics import mean_absolute_percentage_error
y_true = test['Weekly_Sales']
y_pred = forecast
mape = mean_absolute_percentage_error(y_true, y_pred)
print("MAPE:", mape*100)
print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAPE: 7.834223520391406
MAE: 3597509.817290272
RMSE: 3924134.1178595982
```

Forecast Future Sales

In [296…

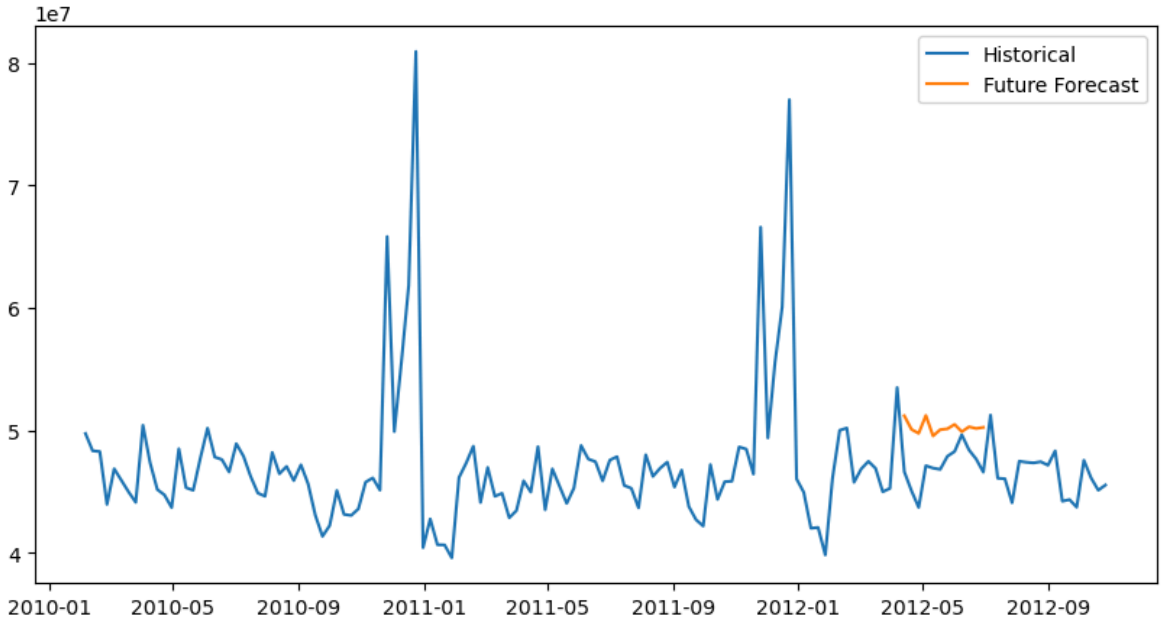
```
future = model_fit.forecast(steps=12)

print(future)
```

```
2012-04-13    5.119490e+07
2012-04-20    5.008505e+07
2012-04-27    4.976354e+07
2012-05-04    5.122423e+07
2012-05-11    4.955298e+07
2012-05-18    5.006257e+07
2012-05-25    5.013423e+07
2012-06-01    5.049133e+07
2012-06-08    4.989652e+07
2012-06-15    5.029623e+07
2012-06-22    5.016284e+07
2012-06-29    5.024870e+07
Freq: W-FRI, Name: predicted_mean, dtype: float64
```

Plot:

In [297…

```
plt.figure(figsize=(10,5))
plt.plot(sales, label="Historical")
plt.plot(future, label="Future Forecast")
plt.legend()
plt.show()
```



Forecasting using SARIMA

In [298…

```
#pip install pmdarima
```

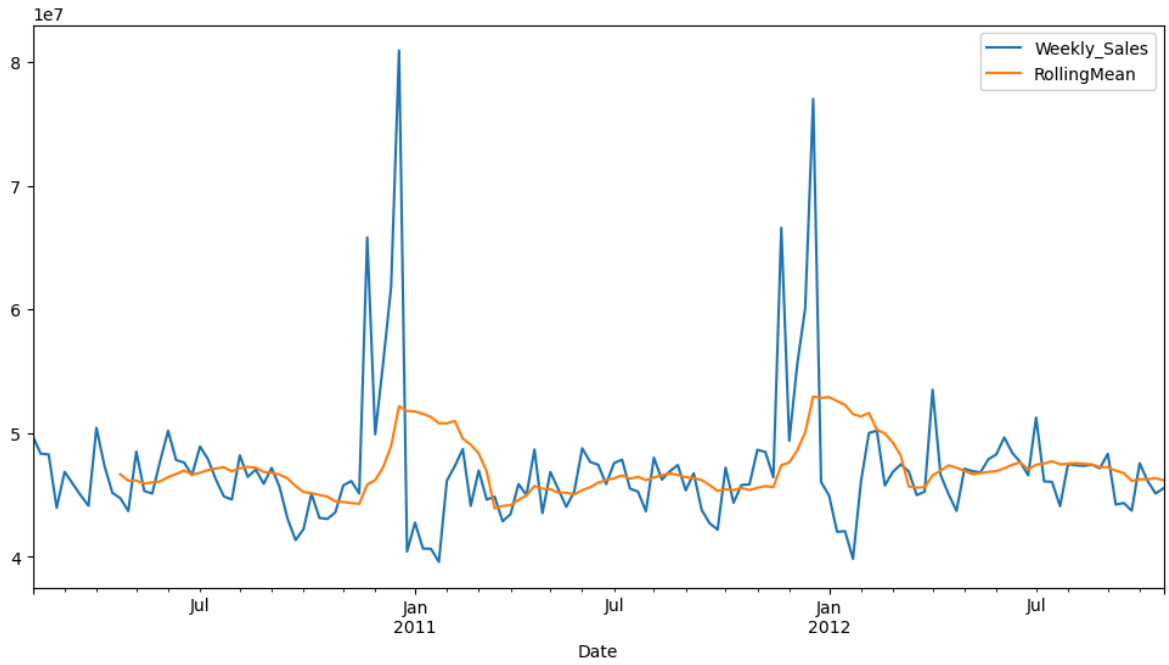
Rolling mean visualization:

In [299…

```
sales['RollingMean'] = sales['Weekly_Sales'].rolling(window=12).mean()

sales[['Weekly_Sales','RollingMean']].plot(figsize=(12,6))
```

Out[299…

```
<Axes: xlabel='Date'>
```



Feature Engineering (Lag Features)

In [300…

```
sales['lag_1'] = sales['Weekly_Sales'].shift(1)
sales['lag_2'] = sales['Weekly_Sales'].shift(2)
sales['lag_3'] = sales['Weekly_Sales'].shift(3)

sales.dropna(inplace=True)
```

auto_arima

In [301…

```
from pmdarima import auto_arima

model = auto_arima(train,
                   seasonal=True,
                   m=12,
                   trace=True)
```

```
Performing stepwise search to minimize aic
 ARIMA(2,0,2)(1,0,1)[12] intercept   : AIC=inf, Time=1.58 sec
 ARIMA(0,0,0)(0,0,0)[12] intercept   : AIC=3886.296, Time=0.03 sec
 ARIMA(1,0,0)(1,0,0)[12] intercept   : AIC=3876.060, Time=0.29 sec
 ARIMA(0,0,1)(0,0,1)[12] intercept   : AIC=3879.935, Time=0.13 sec
 ARIMA(0,0,0)(0,0,0)[12]             : AIC=4356.183, Time=0.03 sec
 ARIMA(1,0,0)(0,0,0)[12] intercept   : AIC=3874.374, Time=0.04 sec
 ARIMA(1,0,0)(0,0,1)[12] intercept   : AIC=3875.883, Time=0.12 sec
 ARIMA(1,0,0)(1,0,1)[12] intercept   : AIC=inf, Time=0.75 sec
 ARIMA(2,0,0)(0,0,0)[12] intercept   : AIC=3874.676, Time=0.07 sec
 ARIMA(1,0,1)(0,0,0)[12] intercept   : AIC=3875.641, Time=0.06 sec
 ARIMA(0,0,1)(0,0,0)[12] intercept   : AIC=3878.647, Time=0.06 sec
 ARIMA(2,0,1)(0,0,0)[12] intercept   : AIC=3876.670, Time=0.12 sec
 ARIMA(1,0,0)(0,0,0)[12]             : AIC=inf, Time=0.03 sec

Best model:  ARIMA(1,0,0)(0,0,0)[12] intercept
Total fit time: 3.354 seconds
```

SARIMAX

In [302…

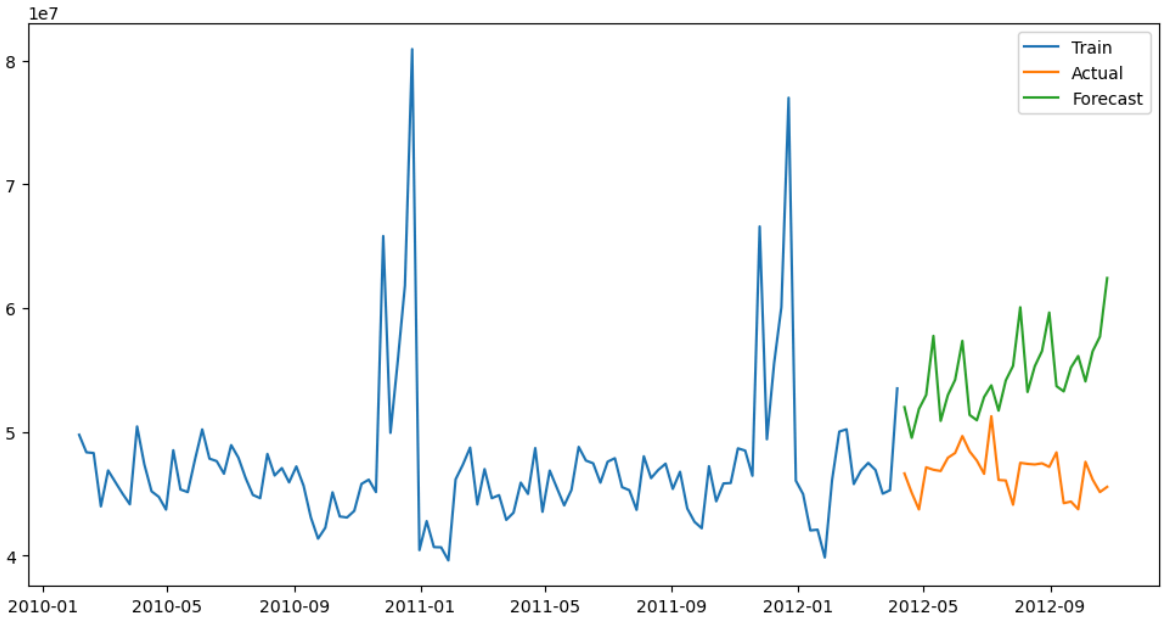
```
from statsmodels.tsa.statespace.sarimax import SARIMAX
model = SARIMAX(train['Weekly_Sales'],
                order=(1,1,1),
                seasonal_order=(1,1,1,12))

result = model.fit()

forecast = result.forecast(steps=len(test))
```

```
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency W-FRI
will be used.
  self._init_dates(dates, freq)
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency W-FRI
will be used.
  self._init_dates(dates, freq)
```

FORECAST

In [303…

```
plt.figure(figsize=(12,6))

plt.plot(train.index, train['Weekly_Sales'], label='Train')
plt.plot(test.index, test['Weekly_Sales'], label='Actual')
plt.plot(test.index, forecast, label='Forecast')

plt.legend()
plt.show()
```



Model Evaluation

In [304…

```
mae = mean_absolute_error(test, forecast)
rmse = np.sqrt(mean_squared_error(test, forecast))
from sklearn.metrics import mean_absolute_percentage_error
y_true = test_sales['Weekly_Sales']
y_pred = forecast
mape = mean_absolute_percentage_error(y_true, y_pred)
print("MAPE:", mape*100)
print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAPE: 17.071662684170036
MAE: 7895017.798036902
RMSE: 8621665.73951517
```

Forecast Future Sales

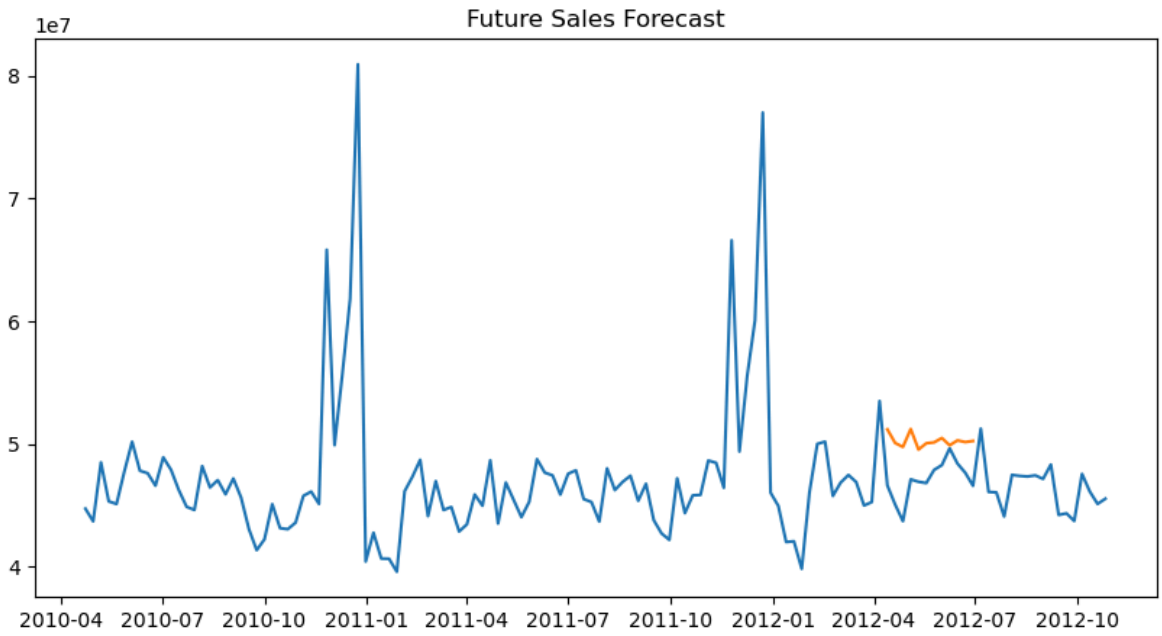
In [305…

```
future_sales = model_fit.forecast(steps=12)

print(future)
```

```
2012-04-13    5.119490e+07
2012-04-20    5.008505e+07
2012-04-27    4.976354e+07
2012-05-04    5.122423e+07
2012-05-11    4.955298e+07
2012-05-18    5.006257e+07
2012-05-25    5.013423e+07
2012-06-01    5.049133e+07
2012-06-08    4.989652e+07
2012-06-15    5.029623e+07
2012-06-22    5.016284e+07
2012-06-29    5.024870e+07
Freq: W-FRI, Name: predicted_mean, dtype: float64
```

In [306…

```
plt.figure(figsize=(10,5))

plt.plot(sales.index, sales['Weekly_Sales'])
plt.plot(future_sales.index, future_sales)

plt.title("Future Sales Forecast")
plt.show()
```



In [307…

```
from sklearn.metrics import mean_absolute_error, mean_squared_error
from statsmodels.tsa.statespace.sarimax import SARIMAX

#from prophet import Prophet
```

SARIMAX modeL WITH exogenous variables

In [308…

```
train= pd.read_csv(r'C:\Users\anupa\Desktop\Python Learning\Forecasting\train.cs
features = pd.read_csv(r"C:\Users\anupa\Desktop\Python Learning\Forecasting\feat
stores = pd.read_csv(r"C:\Users\anupa\Desktop\Python Learning\Forecasting\stores
```

Merge datasets

In [309…

```
df = train.merge(features, on=["Store", "Date", "IsHoliday"])
df = df.merge(stores, on="Store")

df['Date'] = pd.to_datetime(df['Date'])
df = df.sort_values('Date')

df.head()
```

Out[309…

	Store	Dept	Date	Weekly_Sales	IsHoliday	Temperature	Fuel_Price	MarkDo
0	1	1	2010-02-05	24924.50	False	42.31	2.572	
277665	29	5	2010-02-05	15552.08	False	24.36	2.788	
277808	29	6	2010-02-05	3200.22	False	24.36	2.788	
277951	29	7	2010-02-05	10820.05	False	24.36	2.788	
278094	29	8	2010-02-05	20055.64	False	24.36	2.788	

Aggregate Weekly Sales

In [310…

```
#For simplicity we forecast total weekly sales.

sales = df.groupby('Date')['Weekly_Sales'].sum().reset_index()
sales = sales.set_index('Date')
```

Add Exogenous Variables

In [311…

```
exog = df.groupby('Date')[['Temperature','Fuel_Price','CPI','Unemployment']].mea
```

Train Test Split

In [312…

```
train_size = int(len(sales)*0.8)

train_sales = sales[:train_size]
test_sales = sales[train_size:]

train_exog = exog[:train_size]
test_exog = exog[train_size:]
```

Build SARIMAX Model

In [313…

```
from statsmodels.tsa.statespace.sarimax import SARIMAX

model = SARIMAX(train_sales,
                exog=train_exog,
                order=(1,1,1),
                seasonal_order=(1,1,1,52))

results = model.fit()
```

```
print(results.summary())
```

```
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency W-FRI
will be used.
  self._init_dates(dates, freq)
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency W-FRI
will be used.
  self._init_dates(dates, freq)
C:\Users\anupa\anaconda3\Lib\site-packages\statsmodels\tsa\statespace\sarimax.py:
866: UserWarning: Too few observations to estimate starting parameters for season
al ARMA. All parameters except for variances will be set to zeros.
  warn('Too few observations to estimate starting parameters%s.'
                                     SARIMAX Results
==============================================================================
=========
Dep. Variable:                      Weekly_Sales   No. Observations:
114
Model:             SARIMAX(1, 1, 1)x(1, 1, 1, 52)   Log Likelihood
-972.262
Date:                            Thu, 12 Mar 2026   AIC
1962.523
Time:                                    10:56:21   BIC
1981.521
Sample:                                02-05-2010   HQIC
1969.969
                                     - 04-06-2012
Covariance Type:                              opg
==============================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Temperature   5.414e+04   7.67e+04      0.706      0.480   -9.61e+04    2.04e+05
Fuel_Price    3.806e+04   6503.214      5.853      0.000    2.53e+04    5.08e+04
CPI           7.164e+04   4.14e+05      0.173      0.863    -7.4e+05    8.83e+05
Unemployment -6.023e+06   2.21e+04   -272.170      0.000   -6.07e+06   -5.98e+06
ar.L1            0.1859      3.090      0.060      0.952      -5.870       6.241
ma.L1           -0.2326      3.074     -0.076      0.940      -6.258       5.792
ar.S.L52         0.7336      0.136      5.402      0.000       0.467       1.000
ma.S.L52        -0.9419      0.283     -3.329      0.001      -1.496      -0.387
sigma2        4.234e+12      0.082   5.18e+13      0.000    4.23e+12    4.23e+12
==============================================================================
==
Ljung-Box (L1) (Q):                   6.99   Jarque-Bera (JB):                 4.
81
Prob(Q):                              0.01   Prob(JB):                         0.
09
Heteroskedasticity (H):               1.36   Skew:                             0.
44
Prob(H) (two-sided):                  0.50   Kurtosis:                         4.
05
==============================================================================
==

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-st
ep).
[2] Covariance matrix is singular or near-singular, with condition number 5.84e+2
8. Standard errors may be unstable.
```

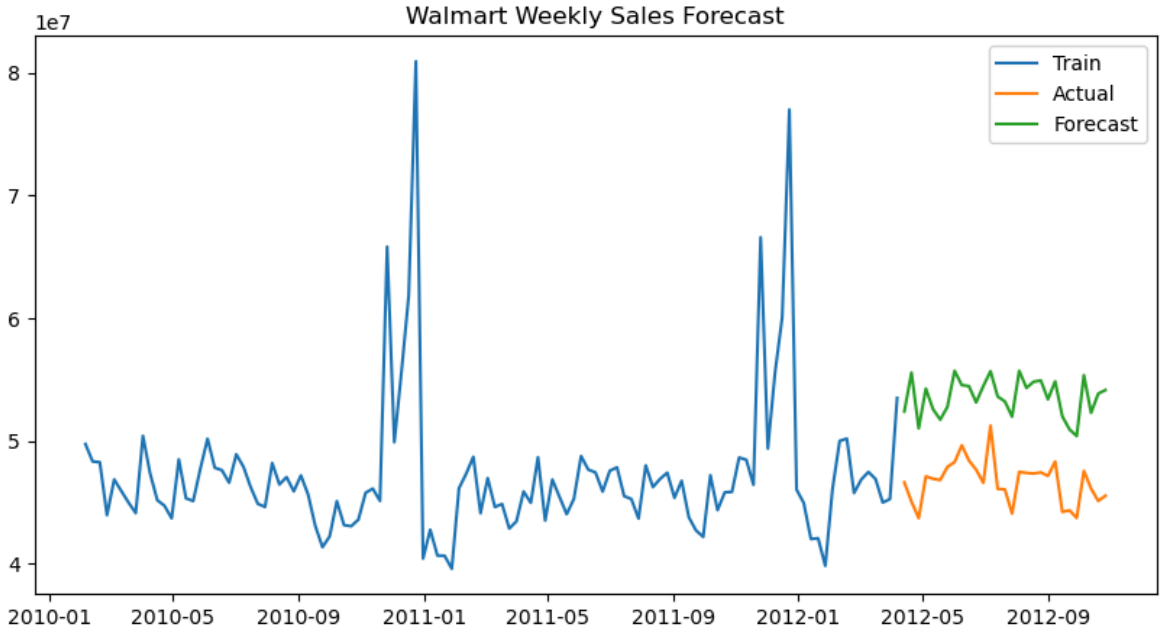
In [314…

```
forecast = results.predict(start=len(train_sales),
                           end=len(sales)-1,
                           exog=test_exog)
```

Plot Forecast vs Actual

In [315…

```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))

plt.plot(train_sales, label="Train")
plt.plot(test_sales, label="Actual")
plt.plot(forecast, label="Forecast")

plt.legend()
plt.title("Walmart Weekly Sales Forecast")
plt.show()
```



Evaluate Forecast

In [316…

```
mae = mean_absolute_error(test_sales, forecast)
rmse = np.sqrt(mean_squared_error(test_sales, forecast))
from sklearn.metrics import mean_absolute_percentage_error
y_true = test_sales['Weekly_Sales']
y_pred = forecast
mape = mean_absolute_percentage_error(y_true, y_pred)
print("MAPE:", mape*100)
print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAPE: 14.862689649460547
MAE: 6904991.032679149
RMSE: 7031216.284046436
```